MILESTONE 2(Data Cleaning & Transformation Implementation)

Objective:

To implement data validation, cleaning, and transformation processes within the ETL pipeline using pandas inside Apache Airflow.

Data Validation & Cleaning Rules Implemented:

Even though dataset was clean, you implemented validation checks:

- Duplicate removal
- Null value detection
- Date format conversion
- Numeric type enforcement
- Column name standardization

Transformation Logic:

Implemented feature engineering:

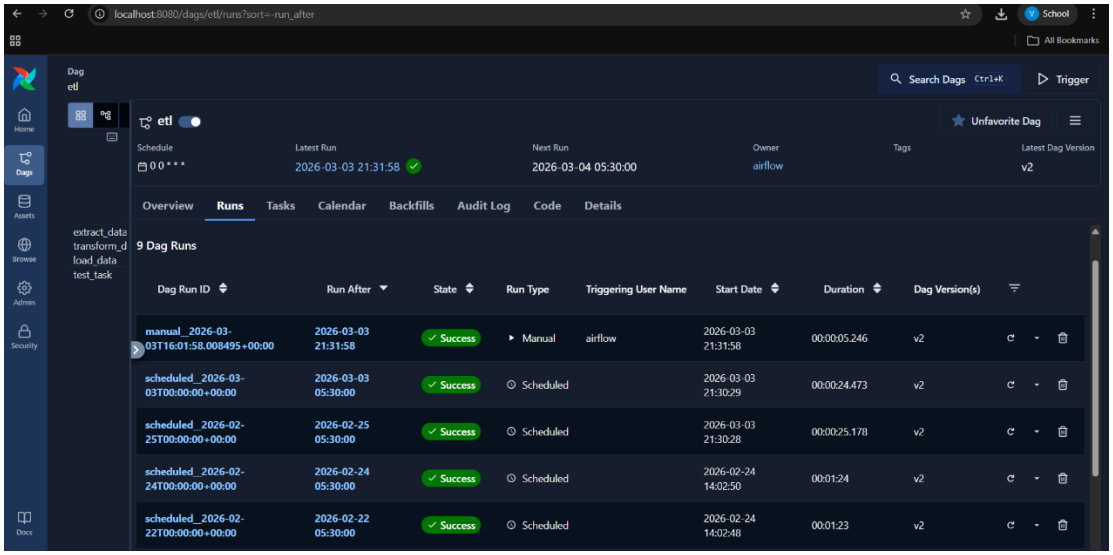
- Converted Date column to datetime
- Converted Profit and TotalPrice to numeric
- Created derived column:

  Profit_Margin = (Profit / TotalPrice) × 100
- Standardized column names (lowercase + underscore)

Logging & Monitoring:

Airflow logs confirm:

- Cleaning process started
- Null value counts
- Duplicate removal
- Final cleaned row count
- Successful completion



Outcome:

- Automated ETL pipeline
- Scheduled execution
- Data quality validation
- Feature engineering completed
- Successful task execution with no failures

etl_dag.py:

```python
from airflow import DAG
from airflow.providers.standard.operators.python import PythonOperator
from datetime import datetime
import pandas as pd

DATA_PATH = "/opt/airflow/data/salesorder.csv"


# -------------------- EXTRACT --------------------
def extract(**context):
    df = pd.read_csv(DATA_PATH, sep="\t")

    print("Raw Rows:", len(df))
    print("Raw Columns:", len(df.columns))

    context["ti"].xcom_push(
        key="raw_data",
        value=df.to_json()
    )


# -------------------- TRANSFORM --------------------
def transform(**context):
    raw_json = context["ti"].xcom_pull(
        task_ids="extract_data",
        key="raw_data"
    )

    df = pd.read_json(raw_json)

    print("Starting Cleaning Process...")

    # 1 Standardize column names
    df.columns = (
        df.columns
        .str.strip()
        .str.lower()
        .str.replace(" ", "_")
    )
```

cleaned_saleorder.csv:

```csv
e,customerid,product,quantity,unitprice,shippingaddress,paymentmethod,orderstatus,trackingnumber,itemsincart,couponcode,referralsource,totalprice,d
023-06-16,C91941,Printer,3,396.0,269 Main St,Gift Card,Cancelled,TRK21330732,4,FREESHIP50,Instagram,1188.0,-50,297.0,50.0,1138.0,25,2023,6
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024-06-05,C71473,Chair,5,3 Col 6: unitprice St,Online,Shipped,TRK94857335,8,FREESHIP50,Instagram,1709.2,-50,427.3,50.0,1659.2,25,2024,6
023-06-12,C62661,Tablet,3,382.37,863 Main St,Debit Card,Returned,TRK25748100,8,No coupon,Instagram,1147.11,0,286.7775,0.0,1147.11,25,2023,6
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024-04-15,C59774,Monitor,1,72.6,941 Main St,Debit Card,Cancelled,TRK87635135,6,FREESHIP50,Instagram,72.6,-50,18.15,50.0,22.6,25,2024,4
023-05-09,C14017,Phone,2,624.16,984 Main St,Debit Card,Shipped,TRK71009446,2,SAVE10,Instagram,1248.32,10,312.08,124.832,1123.488,25,2023,5
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```